

Biopolymers: Structure, Properties and Performance Limits

CM3254 Project Report · [Author 1, Matric No.] · [Author 2, Matric No.] · [Date]

AI Tool Declaration — (final wording to be added on the first page before submission)

1. Introduction and Definition

A biopolymer is a macromolecule synthesised by a living organism: polysaccharides, proteins, nucleic acids, *cis*-1,4-polyisoprene and lignin together account for most of the organic polymer on Earth [1]. The term is extended here to include polymers made by chemical polymerisation of biomass-derived monomers — principally poly(lactic acid) (PLA) — since their properties are best understood alongside the true biopolymers. Three structural features set these chains apart from synthetic polymers: template-directed biosynthesis fixes chain length and sequence, so proteins and nucleic acids are strictly monodisperse ($\bar{D} = 1.00$); biological monomer pools are enantiomerically pure, so chains are stereoregular by default; and hydroxyl/amide/carbonyl groups on almost every repeat unit give cohesive energy densities far above the polyolefins'. This report's properties and limitations follow from these three features: high stiffness and crystallinity, but also brittleness, water sensitivity, and decomposition close to or below the melting point (§5.4).

2. Types of Biopolymers

Biopolymers are often classified first by origin — extracted from biomass, synthesised by micro-organisms, or polymerised from bio-based monomers — but origin predicts behaviour poorly: two materials from the same origin can process and degrade completely differently, while two from different origins can behave alike. The repeating linkage along the backbone is the better predictor, since it determines both melting survival and degradation mechanism, so this review is organised by backbone chemistry instead (Table 1, Fig. 1); origin is retained only as a secondary tag on each material, since it still explains *why* molecular weight and purity vary within a class (§4).

Table 1. Classification of biopolymers by backbone chemistry, the axis this review is organised on; origin (extracted / microbial / bio-monomer) is a secondary property, not the primary split.

Backbone class	Repeating linkage	Representative materials	Characteristic behaviour
Polysaccharide	glycosidic (acetal)	cellulose, starch, chitosan, alginate	dense inter-chain hydrogen bonding; decomposition before melting; enzymatic degradation
Polyester	ester	PHA, PLA	melt-processable but thermally labile; autocatalytic hydrolysis
Protein	amide	silk fibroin, collagen/gelatin	sequence-defined and monodisperse; conformation-controlled properties
Polyphenolic	β -O-4 ether, C–C	lignin	irregular, cross-linked, amorphous; chemically recalcitrant
Polyisoprene	C–C	natural rubber	elastomeric; strain-induced crystallisation

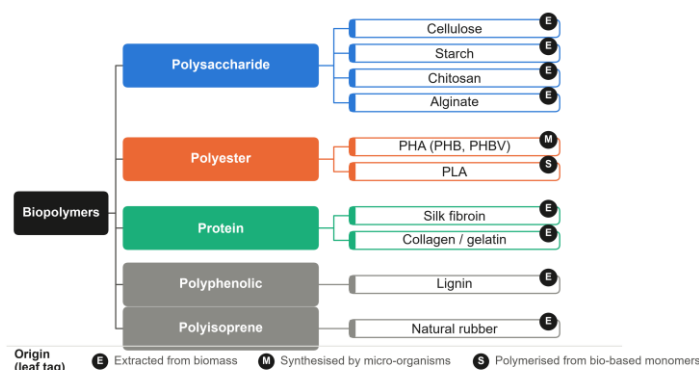


Figure 1. Classification by backbone chemistry (primary branches, coloured, matching Table 1 and Fig. 2), with each material's origin shown as a small E/M/S tag: extracted from biomass, synthesised by micro-organisms, or polymerised from bio-based monomers

3. Structure–Property Relationships of the Major Biopolymer Families

This section examines five biopolymer families along one organising axis: backbone flexibility and the intermolecular forces it permits. Cellulose sits at one extreme — an extended, hydrogen-bonded backbone locked into a crystalline solid that never melts; natural rubber sits at the other — a freely rotating backbone that crystallises only under strain. Between them lie polysaccharides with weaker or ionically tunable interchain forces (starch, chitin/chitosan, alginate), semicrystalline polyesters whose flexibility permits melt processing but only within narrow thermal windows (PHA, PLA), and template-folded proteins whose properties arise from defined secondary/tertiary structure rather than simple chain packing. Lignin — an irregularly cross-linked network with no periodic chain — closes the section as the structural counterpoint to everything else. The claim tested throughout is that backbone chemistry, not biological origin, predicts melting behaviour, hydrolytic/enzymatic susceptibility, and mechanical performance (Table 2). Fig. 2 summarises the force → consequence → limitation causal chain for every family at a glance; the text below adds only mechanism and citations. Each family's monomer and chain-scale structure is drawn directly in its own subsection (§3.1–§3.5); Appendix A (Figs. 18/19) collects the full set in one place.

Family	Intermolecular force	Consequence	Limitation
Cellulose	dense inter- & intrachain H-bonding, sheet stacking	→ very high axial modulus (~130-150 GPa)	→ no melt transition — decomposes at ~300 °C
Starch / chitosan / alginate	weaker, ionically or chemically tunable forces	→ tunable solubility, gelation, charge	→ properties drift with DD, G-content, humidity
PHA (PHB/PHBV)	ester backbone, moderate crystallinity (55-70%)	→ melt-processable phase forms	→ processing window only ~25-55 °C wide
PLA	stereochemistry-controlled chain packing	→ stereocomplex T _m +50 °C over homocrystal	→ T _g ceiling ~60 °C; slow crystallisation
Protein (silk, collagen)	sequence-directed fold, not simple packing	→ combined strength + toughness (silk); triple helix	→ irreversible denaturation collapses structure
Lignin	random radical coupling, no periodic chain	→ chemically recalcitrant aromatic network	→ no single structure-property relationship
Natural rubber	free rotation; strain-induced crystallisation only	→ self-reinforcing elastomer under load	→ MW & network vary by clone and tree age

Colour follows backbone-chemistry class (Table 1); arrows read left to right as one causal chain per family.

Figure 2. Structure-property causal chain for all seven families: dominant intermolecular force → key consequence → key limitation, colour-coded by backbone class

3.1 Polysaccharides

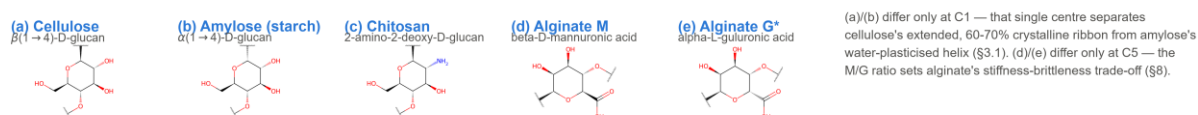


Figure 3. Polysaccharide monomers: cellulose, amylose, chitosan, alginate M/G, drawn from the same verified SMILES as Fig. 18; (a)/(b) differ only at C1, (d)/(e) only at C5

Polysaccharides span the rigid end of the spectrum. **Cellulose** — linear β -(1→4)-D-glucopyranose — forms an extended ribbon locked by hydrogen bonding into sheets (I α , I β); processing is restricted to solution routes (NMMO/Lyocell, LiCl/DMAc), and nanocrystals/nanofibrils exploit its crystal modulus as fillers [2].

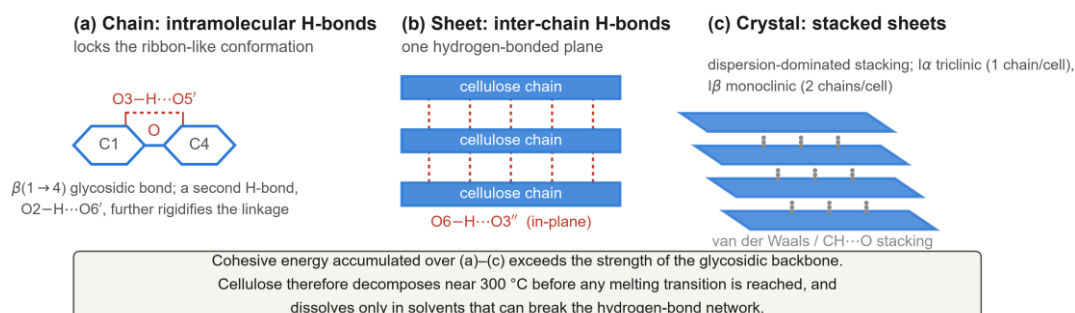


Figure 4. Cellulose's three-tier structure: intrachain H-bond → interchain sheet → interlayer stacking, with a closing line connecting this hierarchy to the 300 °C decomposition-before-melting result

Starch's amylose/amylopectin topology (not chemistry) explains its weaker, humidity-reversible crystallinity (TPS blends with PLA/PBAT). **Chitosan's** degree of deacetylation (DD) sets solubility and antimicrobial activity, but higher-DD chitosan is typically lower-MW since the same alkaline step that raises DD cleaves the backbone [3]. **Alginate** gels via Ca²⁺-coordinated "egg-box" GG-blocks [4], G-content setting the stiffness–brittleness trade-

off (§8). **Others** — hyaluronic acid, carrageenan, xanthan, pectin — follow the same linkage-chemistry logic (§8).

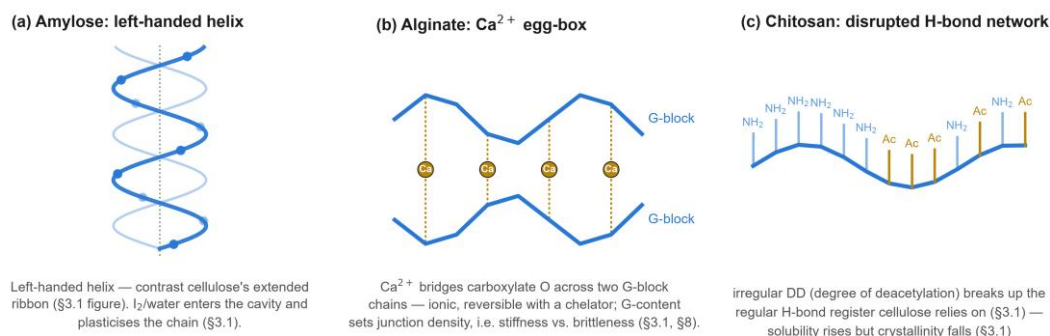


Figure 5. Amylose's left-handed helix, alginate's Ca^{2+} egg-box junction, and chitosan's acetylation pattern breaking up the regular H-bond network, three panels; cellulose's chain-scale structure is already covered in Fig. 4 and not repeated here

3.2 Polyesters: PHA and PLA

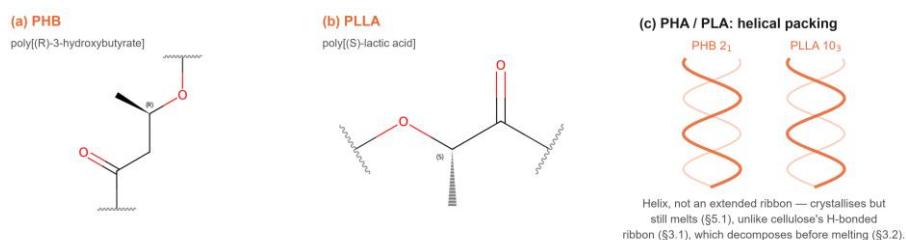


Figure 6. PHB and PLLA repeat units, and PHA/PLA's helical packing ($2_1/10_3$ helices), drawn from the same verified SMILES as Fig. 18

PHA (bacterial storage granules; PHB the archetype) is highly crystalline (55–70%) but decomposes via six-membered-ring *cis*-elimination only tens of degrees above T_m [5] — the narrowest processing window in this report; PHBV/PHBHHx copolymers widen it at the cost of stiffness. **PLA**: ring-opening polymerisation of lactide ($Sn(Oct)_2$) reaches useful MW where direct polycondensation cannot; PLLA crystallises slowly, but the PLLA/PDLA stereocomplex co-crystallises with $T_m \sim 50^\circ C$ above either homocrystal via tighter enantiomeric packing [6,7]. PBS/PEF follow the same ester logic (§8).

3.3 Protein-based Polymers

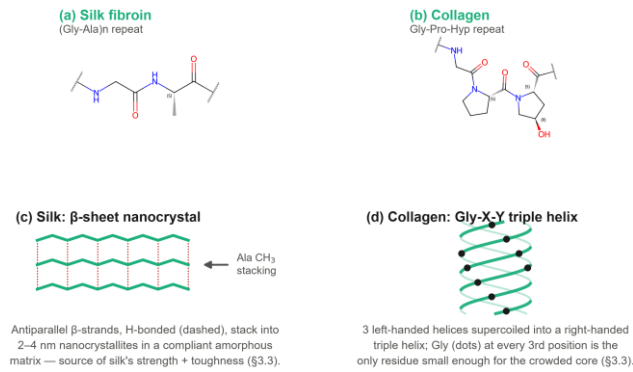


Figure 7. Silk fibroin's (Gly-Ala)_n and collagen's Gly-Pro-Hyp backbone repeat units, and each material's secondary structure: silk's β -sheet nanocrystallite, collagen's triple helix

Protein properties come from sequence-directed fold, not chain packing. **Silk fibroin's** β -sheet nanocrystallites (2–4 nm, required for its combined strength and toughness [8]) sit in a compliant amorphous matrix. **Collagen's** Gly-X-Y triple helix, stabilised by hydroxyproline, denatures irreversibly into gelatin. Zein/casein/soy are film-forming, not structural (§8). Ribosomal synthesis fixes every protein molecule's length and sequence exactly ($\mathfrak{D} = 1.0$) — unmatched by any non-templated biopolymer (§4).

3.4 Lignin and Other Aromatics

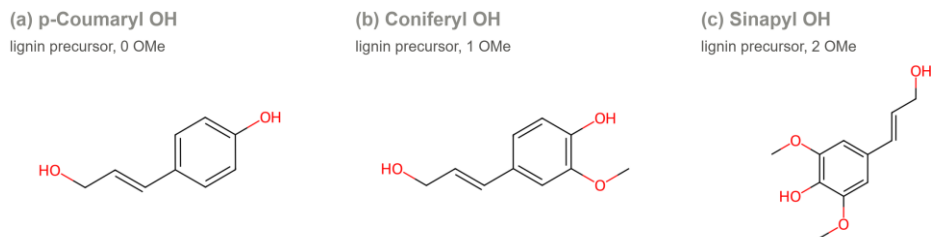


Figure 8. The three monolignol precursors: *p*-coumaryl alcohol (0 methoxy groups), coniferyl alcohol (1), sinapyl alcohol (2)

Lignin is cellulose's counterpoint: rigid *because* regular vs. amorphous *because* irregular. Radical-coupled monolignols give a randomly cross-linked network (β -O-4 ether ~45–60% of linkages [9]) whose composition varies by species and — for industrial use — extraction method (kraft, organosolv, liginosulfonate), which is why no single "lignin" has one structure–property relationship. Lignin has no periodic chain, so only the monomers are drawn here — unlike the other families, an irregularly cross-linked network has no single "typical chain" to picture.

3.5 Natural Rubber

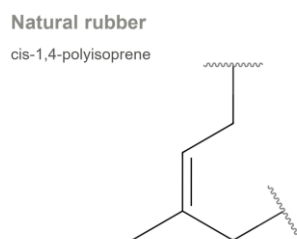
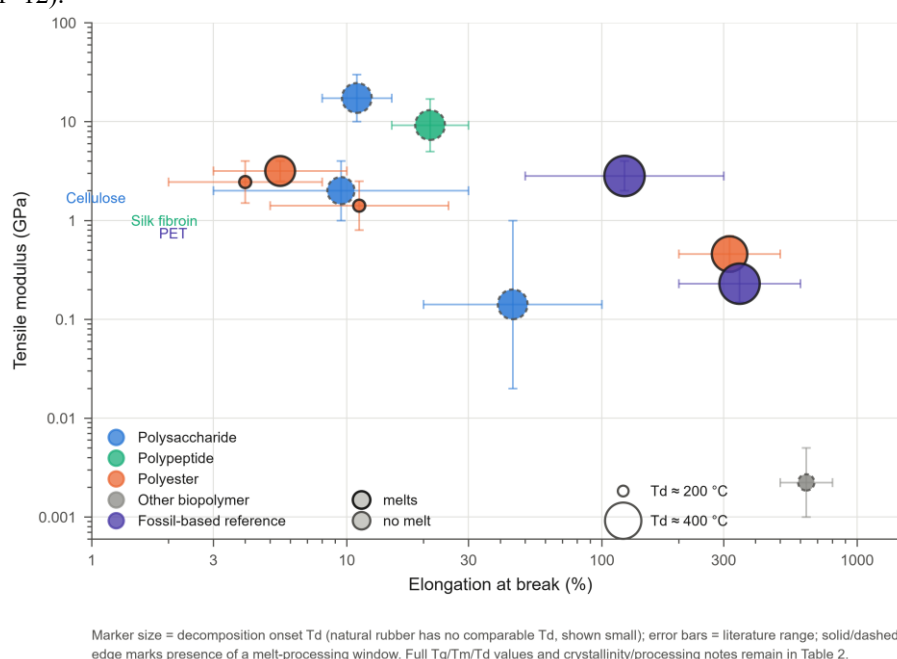


Figure 9. Natural rubber's monomer: *cis*-1,4-polyisoprene repeat unit; the chain-scale strain-crystallisation structure is in §7.3's Fig. 17 and not repeated here

Natural rubber is the flexible extreme and this section's one exception to "crystallinity governs performance": *cis*-1,4-polyisoprene is amorphous at rest, but strain-induced crystallisation (SIC) self-reinforces the network under load — well documented by X-ray diffraction, largely absent in synthetic *cis*-polyisoprene [10]. This is not fixed: NR's molecular weight and distribution vary measurably between *Hevea* clones and with tree age [19], and non-rubber protein/gel content is an active, clone-dependent contributor to network structure — not an incidental impurity [18]. The organising variable here is therefore not "crystallinity" per se but *intermolecular architecture under the conditions of use* — tested directly against a synthetic analogue in §7.3.

Table 2. Provisional summary of thermal, mechanical and structural properties of the representative biopolymers discussed in §3 (order-of-magnitude ranges, cross-checked against the comparative review of biobased thermoplastics by de Beukelaer et al. [11] and, for PBS specifically, Aliotta et al. [12]; source data in *figures/data/*.csv*; several individual *ref* entries still need a material-specific citation before submission — see Fig. 11–12).



All values PROVISIONAL — teaching-level ranges used to make Fig. 11/12 and this table internally consistent; must be replaced with cited literature values before submission.

4. Molecular Weight and Its Measurement

Molecular weight is reported casually in most surveys — "high molecular weight" — but the *distribution*, not just the average, is where the biology shows through. Mn, Mw, dispersity $\bar{D} = Mw/Mn$, and degree of polymerisation (DP) are standard descriptors; $\bar{D}$ matters because it, not Mn alone, determines processing behaviour and the breadth of mechanical property distribution within a batch. **Dispersity is a readout of synthesis mechanism, not noise** (Fig. 10): template-controlled biosynthesis gives proteins and nucleic acids $\bar{D} = 1.0$ exactly, while every non-templated route — extraction (cellulose, chitosan, natural rubber) or catalysed polymerisation (PHA, PLA) — broadens the distribution to a degree set by how tightly that route is controlled. Natural rubber is the extreme case: because it is extracted rather than chain-grown, Mw and MWD vary between *Hevea* clones and with tree age [19] — an agricultural variability with no synthetic-polymer analogue.

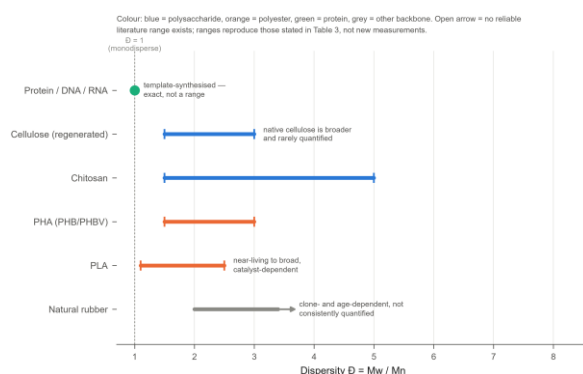


Figure 10. Dispersity $\bar{D}$ across families on a common axis: a point at $\bar{D}=1$ for templated biosynthesis, ranges for PLA/PHA/chitosan/cellulose, an open-ended arrow for natural rubber's unquantified variability

Why molecular weight controls performance. Mechanical integrity depends on chain entanglement above a characteristic M_c ; below it, strength and toughness collapse. Above M_c , tensile strength rises with M_n then saturates (Flory-type, $\sigma \approx \sigma_\infty - K/M_n$), so degradation — first visible as falling MW — can proceed with only modest property loss until crossing back below M_c , where performance collapses abruptly, a failure mode relevant to §7–§8. **Measuring** biopolymer MW is harder than for synthetic polymers: cellulose dissolves in no standard SEC eluent, requiring derivatisation or specialised solvents that can themselves degrade or aggregate the sample, while aggregation and branching distort SEC calibrated against linear standards. SEC-MALS or Mark–Houwink viscometry are more defensible, but Mark–Houwink parameters are missing for many biopolymer/solvent pairs. The result: reported MW for the same material commonly differs by an order of magnitude across the literature — an artefact of method more than genuine variation, so meaningful comparison requires the method too.

Table 3. Typical number-average molecular weight for representative biopolymers, with the standard method by which each is measured (dispersity $\bar{D}$ is plotted in Fig. 10 instead of repeated here).

Material	Typical Mn	Standard method (key limitation)
Proteins / DNA / RNA	sequence-specific	mass spectrometry (none — templated)
Cellulose (native/Lyocell)	10^4 – 10^6 g mol ⁻¹	viscometry/SEC-MALS after derivatisation (insoluble in std. eluents)
Chitosan	10^4 – 10^6 g mol ⁻¹	intrinsic viscosity (DD/MW trade-off, §3.1)
PHA (PHB/PHBV)	10^5 – 10^6 g mol ⁻¹	GPC/SEC vs polystyrene (calibration mismatch)
PLA	10^4 – 10^5 g mol ⁻¹	GPC/SEC (moisture-driven hydrolysis in process)
Natural rubber	often $>10^6$ g mol ⁻¹	intrinsic viscosity (no synthetic analogue)

Ranges are teaching-level approximations; VERIFY each row against a primary source and replace with cited values before submission.

5. General Property Patterns

Section 3 examined each family in turn; this section turns the same evidence sideways, asking what pattern holds across all of them.

5.1 Thermal Behaviour

A decomposition temperature at or below the melting point is close to the rule here (Fig. 11): the same network that gives stiffness also raises the melting energy toward the bond-breaking energy, crowding the two transitions together — cellulose, chitosan and silk fibroin never melt at all. Among the polyesters this crowding is quantitative: PHB's window spans only ~25 °C, PHBV widens it to ~55 °C, and PLLA's is wider still (§3.2). Fig. 11 plots the TGA onset conservatively — for PLLA and PHB the usable window is narrower still, since melt hydrolysis (§4) begins earlier.

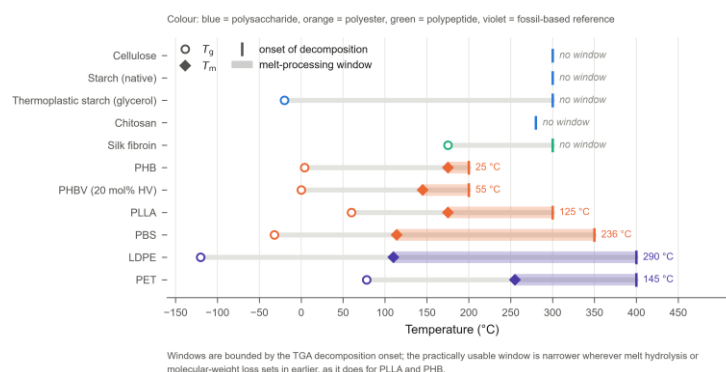


Figure 11. Thermal properties and processing windows: Tg/Tm/Td for each material on a common temperature axis, window width labelled directly

5.2 Mechanical Behaviour

The same logic sets the mechanical envelope (Fig. 12): high crystallinity and dense hydrogen bonding raise modulus but remove the flexible segments that let polyolefins draw before breaking, so biopolymers cluster toward high modulus, low elongation relative to PE/PP/PET. Natural rubber is the exception — no modulus at rest, performance generated on demand by SIC (§3.5, §7.3). Fig. 12's windows are regions, not points: sample form, moisture,

crystallinity and thermal history each shift a material's data by an order of magnitude, recurring as "batch variability" in §7.2.

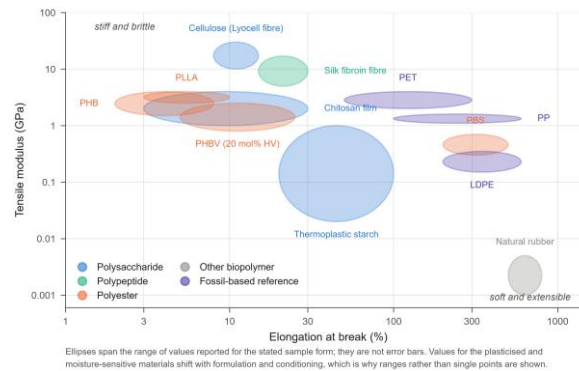


Figure 12. Ashby-style modulus vs. elongation-at-break map, region ellipses in log–log space, PE/PP/PET plotted for reference

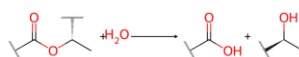
5.3 Hydrophilicity and Barrier Properties

The polar groups behind §5.1–5.2's strength have a third consequence: strong hygroscopicity. Absorbed water plasticises the amorphous fraction and swells free volume, so barrier performance falls as humidity rises. ⚙️ Atomistic free-volume/diffusion estimates offer a route to predicting this computationally, ahead of a packaging trial — which is why PLA and starch films must be specified with a storage-humidity condition, not a fixed barrier rating.

5.4 Degradation as a Material Property

Degradation is a rate process governed by structure (Figs. 13–15). Polyesters degrade by autocatalytic ester hydrolysis (each scission's acid end catalyses further hydrolysis, so interiors degrade faster than surfaces; Fig. 13); polysaccharides and proteins instead degrade by enzymatic attack, which can only reach amorphous segments, so rate is set by accessible surface area, not bulk chemistry (Fig. 14). Crystallinity therefore recurs as this report's central trade-off — the same feature that sets modulus (§5.2) and blocks water uptake (§5.3) also blocks the attack that would degrade the material. ⚙️ Computed barrier heights for neutral/acid/ base-catalysed hydrolysis reproduce the qualitative rate ordering in Fig. 15, which is illustrative only.

Polyesters: backbone hydrolysis



Acid end product autocatalyses further hydrolysis — bulk erosion.

Figure 13. Polyester backbone ester hydrolysis mechanism, autocatalytic bulk erosion

Polysaccharides and proteins: enzymatic attack



Enzymes reach only amorphous regions — crystallinity sets the rate.

Figure 14. Polysaccharide/protein enzymatic degradation, limited to amorphous regions

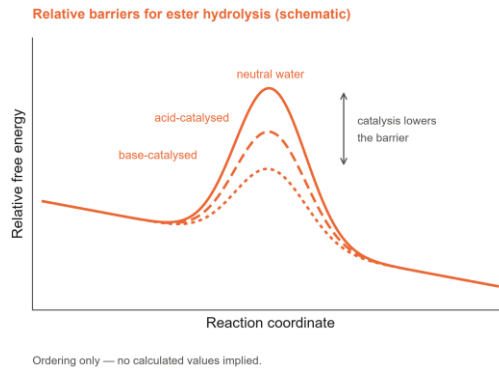


Figure 15. Qualitative energy-barrier ordering for neutral/acid/base-catalysed ester hydrolysis, no values implied

6. Processing

Fig. 16 turns §5.1's windows into a processing decision: whether a stable melt window exists at all sets melt vs. solution processing, and each material's window width then sets how tightly that process must be controlled. The recurring judgement this yields: most reported "performance shortfalls" are processing-window shortfalls, not material-property shortfalls (§5.1–5.2).

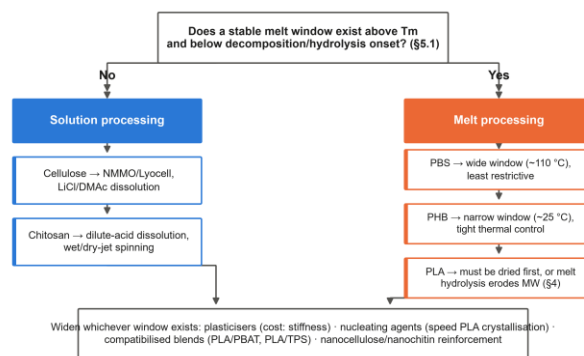


Figure 16. Processing decision flowchart: stable melt window? → melt processing (PBS/PHB/PLA, window width sets control tightness) or solution processing (cellulose/chitosan dissolution routes) → shared countermeasures that widen whichever window exists

7. Advantages and Disadvantages Compared with Synthetic Polymers

7.1 Where Biopolymers Genuinely Win

Four advantages are structural, each tracing back to §3: **renewable, non-fossil feedstock** — the only one applying uniformly across all families; **biodegradability in leakage-prone**

contexts (mulch film, fishing gear, disposable foodware), where hydrolytic/enzymatic susceptibility becomes desired if the environment matches the material (§5.4); **biocompatibility/bioresorbability** — PLA/PGA sutures, collagen/alginate hydrogels, chitosan dressings do what commodity synthetics cannot without costly modification, the one space (§8) where biopolymers are the only viable class, though not unconditionally (§7.3); and **intrinsic functionality beyond mechanics** — chirality (PLA stereocomplexation), pH-responsive charge (chitosan), sequence-encoded bioactivity (silk, collagen), all free from biology.

7.2 Where They Lose — Stated with Numbers, Not Adjectives

(Order-of-magnitude ranges consistent with cited values; each still needs a tighter primary citation before submission.)

- **Price.** PE/PP ~\$1–1.5/kg; PLA several times higher; PHA typically priciest. [*cite market report*]
- **Heat/barrier.** PLA's HDT (~55–65 °C) sits well below PP's ~100 °C+ without stereocomplexation (§3.2) [13]; its O₂/CO₂ transmission is comparable to PET's, but water-vapour barrier is worse, and starch films worse again [14] — the correct critique is moisture sensitivity (§5.3), not a blanket oxygen deficit.
- **Brittleness/hygroscopicity.** PHB's secondary crystallisation and PLA's sub-T_g ageing embrittle over storage; starch drifts with humidity (§3.1, §5.3) and PLA must be dried before melt processing to avoid uncontrolled MW loss (§4).
- **Batch variability.** Agricultural sourcing (alginate M/G, chitosan DD, starch amylose, NR clone/age [19]) varies in ways petrochemical monomers do not.
- **Recycling compatibility.** A 2022 study found in-stream PLA at only 0–0.019% of PET recycling streams, with NIR sorting keeping it below the ~1% threshold that measurably affects rPET [15] — a sorting-infrastructure dependency, not an inherent incompatibility.

7.3 Case Study: Natural Rubber vs. a Synthetic Analogue (SBR)

NR and styrene–butadiene rubber (SBR) are compounded, filled and vulcanised the same way and compete directly in the same applications (tyre tread, damping, conveyor belting) — the cleanest same-application comparison here, since differences trace to backbone chemistry, not processing. Fig. 17 draws out the chain-scale origin of that difference (SIC; full comparison in Fig. 19(f)); Table 5 then scores both across seven dimensions, with citation numbers for every row, expanded in the References [10,18–32].

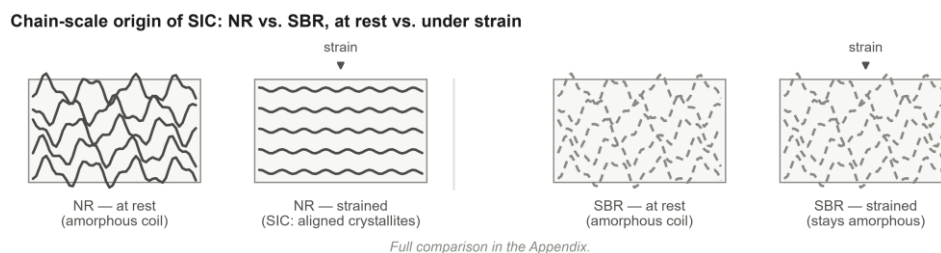


Figure 17. Chain-scale structure of NR and SBR at rest vs. under strain, a direct picture of SIC; full version also in Appendix Fig. 19(f)

Table 5. Natural rubber (NR) vs. SBR, dimension by dimension — structurally favourable / formulation- or context-dependent / documented risk or caveat.

Dimension	Natural rubber (NR)	SBR
Structure / MW control	Context-dependent: extracted; varies by clone & tree age [19]	Favourable: engineered via emulsion/solution copolymerisation [20,21]
SIC & tear resistance	Favourable: strong strain-induced crystallisation [10,23]	Context-dependent: crystallises far less; relies on filler/cure [24]
Blend performance	Context-dependent: SIC weakens as SBR fraction rises [22]	Context-dependent: reaches parity via reinforcement, not backbone [24]
Tg / cure network	Context-dependent: depends on formulation & cure conditions [25,26]	Context-dependent: depends on formulation & cure conditions [25,26]
Biocompatibility	Context-dependent: real potential, but latex protein allergy risk [27,28]	Risk: no unconditional biocompatibility claim [27,28]
Environmental footprint	Context-dependent: renewable, but LCA shows real land/energy cost [29,30]	Risk: fossil feedstock; biodegradable grades emerging [31]
Price stability	Risk: historically the more volatile of the two [32]	Context-dependent: steadier, but tied to petrochemical supply chain [32]

NR is extracted, so its structure and SIC contribution are agricultural variables (clone, tree age); SBR's composition is instead engineered by emulsion/solution copolymerisation in a continuous industrial process [20,21]. Under matched compounding, NR's SIC gives superior tear and crack-growth resistance to start with [10,23], but SBR reaches comparable wear resistance through reinforcement and cure design rather than backbone crystallisation [24] — the clearest single illustration here of backbone chemistry trading off against formulation (§6). The "natural is better" story does not survive contact with the literature, though: NR latex carries a real, SBR-absent allergy risk [27,28], its renewability does not translate into a lower LCA footprint automatically [29,30], and NR pricing has historically been the more volatile of the two for climate/disease reasons unrelated to polymer chemistry [32].

8. Applications

Each application states the property requirement first, then names the material that satisfies it (Table 4). **Packaging** needs O₂/H₂O barrier, toughness, and heat resistance exceeding PLA's HDT (§7.2); no single biopolymer satisfies all three (§5.3), so packaging is almost always multilayer or blended. Compostable foodware and mulch film are where §7.1's biodegradability, not barrier, drives the design, needing a rate matched to use timescale (§5.4). **Biomedical** applications need biocompatibility plus a tunable degradation rate (§5.4) — essentially unique here (§7.1, with the NR-latex caveat in §7.3): PGA/PLGA are established

suture materials whose main failure mode is local pH drop from hydrolysis products; collagen/alginate/chitosan extend the logic to scaffolds and bioprinting. **Textiles/other:** Lyocell and PLA fibre exploit cellulose's solution-processed stiffness (§3.1, §6) and PLA's melt-spinnability (§3.2); NR adhesives and PHA/PLA electronics substrates trade predictability for intrinsic adhesion, biodegradability, or biocompatibility.

Table 4. Applications, required property, and material.

Application	Key property required	Representative material(s)
Food packaging	O ₂ /H ₂ O barrier + toughness + moderate heat resistance	PLA/PBAT blends; PHA/PLA coatings
Compostable foodware / mulch film	Biodegradability matched to use timescale, not barrier	PLA, starch blends, PBAT/PLA
Sutures / drug release / scaffolds	Biocompatibility + tunable degradation rate	PGA, PLGA, collagen, alginate, chitosan
Textile fibre / electronics substrate	High stiffness/spinnability or film formability	Lyocell, PLA fibre, PHA
Tyre tread / damping / conveyor belting	High tensile/tear strength via SIC, or engineered wear	Natural rubber, NR/SBR blends (§7.3)

9. Current Status, Challenges and Outlook

Global bio-based plastics capacity was ~2.31 million tonnes in 2025 — under 1% of global plastics — projected to double by 2030 [16]. Growth concentrates in bio-based PE/PP and PHA, expanding fastest despite remaining priciest (§7.2) [16]. The bottlenecks are the same three properties organising §5 — processing windows, humidity-dependent performance, batch variability — none solved by scale alone. Two directions look most promising: enzymatic depolymerisation back to monomer (recent enzyme cocktails reached ~60% lactic-acid recovery from post-consumer PLA within 72 hours [17]); and ⚙ data-driven property prediction (QSPR/ML) to shorten the trial-and-error cycle of matching backbone chemistry to a processing window.

10. Conclusions

Three judgements follow. First, the performance ceiling here is set by thermal and hygroscopic behaviour (§5.1, §5.3), not mechanical strength (§5.2) — most families meet or exceed synthetic-polymer stiffness and fail instead on processing window and moisture sensitivity. Second, crystallinity governs the entire trade-off (§5.4): the same feature that supplies stiffness, blocks water uptake, and blocks the attack that would degrade the material means raising one predictably lowers the other two — the NR/SBR case study (§7.3) shows the same trade-off through formulation, not backbone chemistry alone. Third, molecular weight data are only comparable alongside their measurement method (§4): the features that make it hard to measure — hydrogen bonding, branching, sequence specificity, agricultural sourcing — are the same features used throughout to explain performance.

References ***(32 verified; assignment requires 50–60 — see REFERENCES-TODO below)***

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Appendix A: Monomer and Chain-Scale Structures

§3.1–§3.5 now each carry their own compact monomer and chain-scale structure figures (Figs. 3/4/5/6/7/8/9), placed right where the argument needs them. This appendix is not a new argument — it collects every material (including SBR, which appears only in §7.3 and gets no standalone monomer figure there) into two complete reference figures in one place — like the References, it does **not** count against the 10-page body limit (OUTLINE.md: "10 页，不含参考文献").

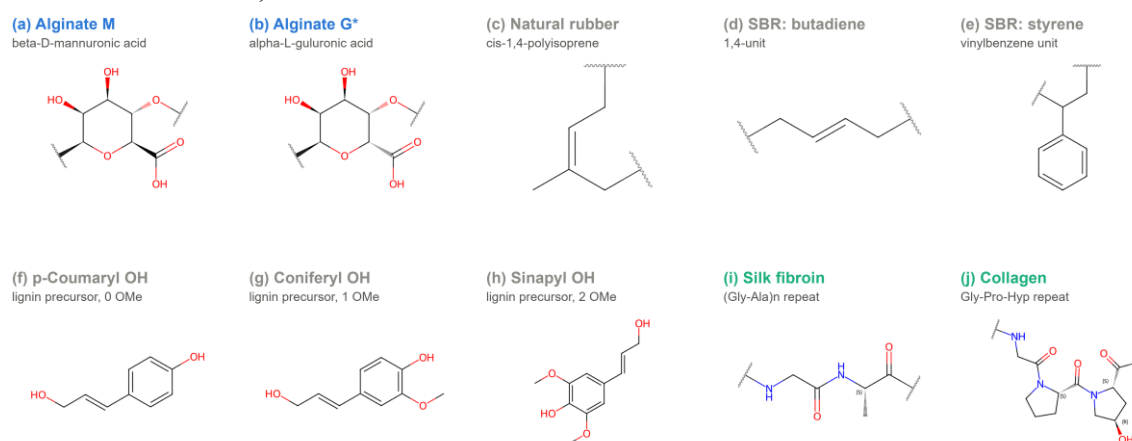


Figure 18. Repeat units for fifteen materials: cellulose, amylose, chitosan, PHB, PLLA (as Figs. 3/6), alginate M/G (as Fig. 3), natural rubber (as Fig. 9), SBR's two comonomers (appears only here — §7.3 does not draw SBR's monomer separately), the three monolignols (as Fig. 8), and the silk/collagen backbones (as Fig. 7), drawn by RDKit from verified SMILES. Alginate G (guluronic acid)'s absolute configuration is verified only as "the C5 epimer of M" (a literature-established relationship) — no independent literature CIP string was available to cross-check it directly; check against ChemDraw/a literature structure before submission, see the header of `figures/verify_appendix_stereochemistry.py``

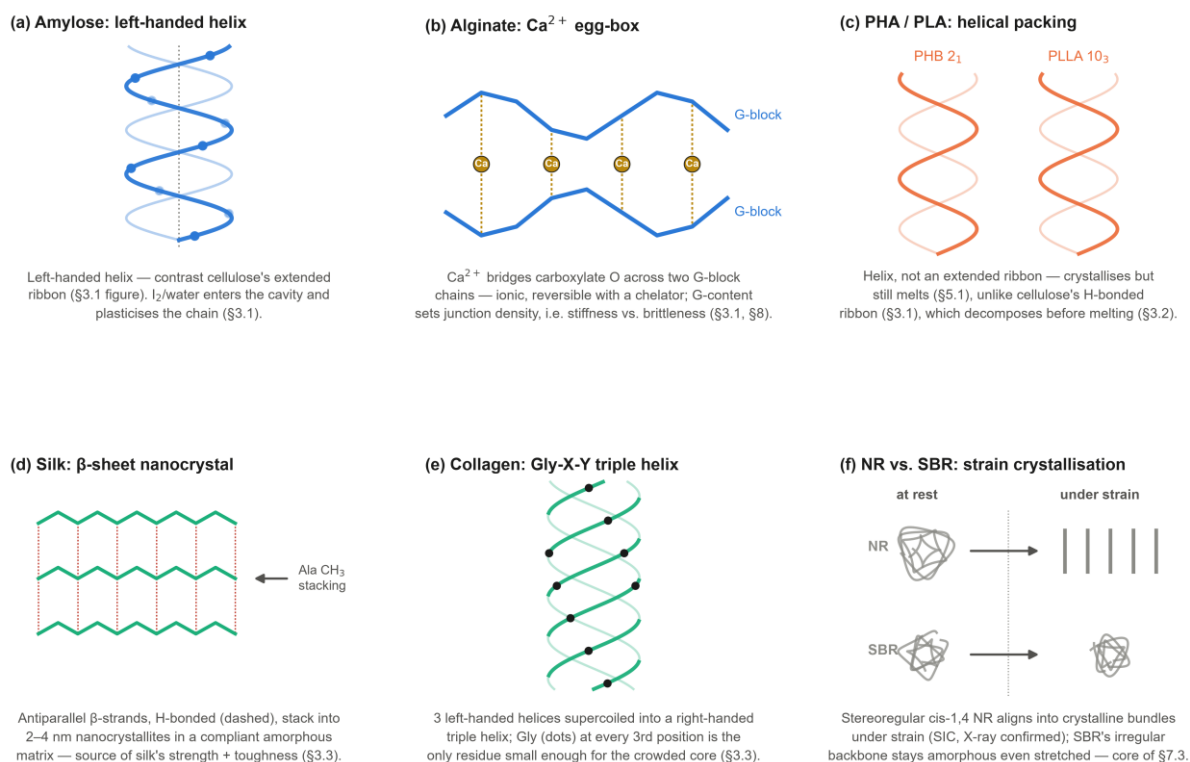


Figure 19. Chain-scale spatial structure for six families, collected: (a) amylose's left-handed helix (as Fig. 5a) (b) alginate's Ca^{2+} "egg-box" junction (as Fig. 5b) (c) PHA/PLA helical packing, $2_1/10_3$ helices (as Fig. 6c) (d) silk fibroin's β -sheet nanocrystallites (as Fig. 7c) (e) collagen's Gly-X-Y triple helix (as Fig. 7d) (f) natural rubber vs. SBR under strain, the structural core of the §7.3 case study — a compact version already sits in Fig. 17 next to the argument it supports, this is the full version; cellulose's chain-scale structure is already covered in its own three-panel figure (Fig. 4) and not repeated here